The Harsh Realities of Parallelization

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The harsh realities of parallelization (1)

- In an ideal world, a uniprocessor vs an n-way multiprocessor -> n-fold increase in computational power.

- In practice, this (almost) never happens -> due to the costs of interprocessor communication & coordination.

- E.g., 5 friends painting a 5-room house. Assume that all the rooms are the same size, everyone is assigned to each room only and paints at about the same rate -> ?-fold speedup?!

- But what if the rooms are different sizes?! -> the overall completion time is dominated by one room that takes the longest to paint.

The harsh realities of parallelization (2)

- Speedup S

$$S = \frac{The\ time\ it\ takes\ one\ processor\ to\ complete\ the\ job}{The\ time\ it\ takes\ n\ concurrent\ processors\ to\ complete\ the\ job}$$

- Let p be the fraction of the job that can be executed in parallel. Amdahl's Law:

$$S = \frac{1}{(1-p) + \frac{p}{n}}$$

- Assume that each small room is one unit, and the single large room is two units -> $S = 3$
- How about 10 rooms & 10 painters?! -> $S = 5.5$
- What does Amdahl's law imply?! -> Even if we manage to parallelize 90% of the solution, but not the remaining 10%, then we end up with a 5-fold speedup, not a 10-fold speedup.

➔ The remaining 10% that we didn't parallelize cut our utilization of the machine in half!!!

The harsh realities of parallelization (3)

- It seems worthwhile to invest effort to derive as much parallelism from the remaining 10% as possible, even if it is difficult.
- Question: Any solutions to improve the remaining 10%?
- Solution: As soon as one painter's work is done, that painter helps others to paint the remaining room.
- But how can this shared painting of the room be done? -> Requiring some communication and coordination.
- Typically, it is hard because these additional parallel parts involve substantial communication and coordination.